

## Trainings ▾

### General Information

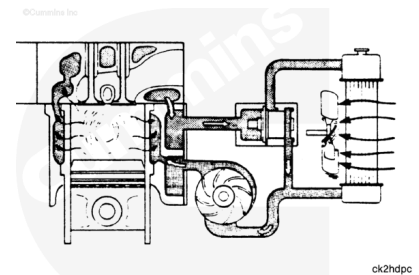
#### Injection Cycle

##### Water Pump With Exhaust Gas Recirculation (EGR)

On the 2002 ISM engine, the water pump is built with an impeller similar to that used on the QSK19 engine to maximize the cooling capacity of the water pump. The water pump pulley drive ratio was increased to provide faster pump speeds to increase the cooling capacity. Increased cooling capacity is required due to the extra flow and heat associated with EGR.

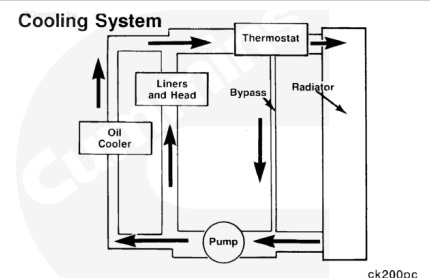
The primary function of the cooling system is to remove the heat created by the engine and its support components. The excess heat energy that is **not** removed by the cooling system is carried away by exhaust gases and radiation into the atmosphere.

The coolant **must** be made up of proper proportions of water, antifreeze and supplemental coolant additives (SCA) to properly perform these functions.



The chart in the graphic illustrates the coolant flow through the engine. For more detail, see the coolant flow diagrams.

- Charge air cooled engines



The following publications, available through a Cummins® Authorized Repair Location, provide cooling system installation recommendations and specifications approved by Cummins Inc.:

- Data sheets for specific engine models.

